

Eric Wu

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Education

Stony Brook University

Program: *College of Engineering and Applied Sciences*

Major: *Computer Engineering B.E.*

Stony Brook, NY

August 2022 - May 2026

Current GPA: **3.5**

Relevant Courses: Embedded Microprocessor Systems Design, Digital Systems Design, Non-Linear Electronics, Advanced Programming & Data Structures, Real Time Operating Systems, Computer Architecture, Senior Design Project, PCB Design

Skills

Programming Languages: C, C++, MIPS Assembly, Embedded Assembly, Embedded C, VHDL, SystemVerilog, MATLAB, Python

CAD: Autodesk Inventor, Autodesk Fusion 360, Cadence OrCAD, LTSpice, Altium

Operating Systems: Windows, macOS, Unix/Linux

Suites: Google Cloud Platform, Google Workspace, Microsoft 365

Hardware: Oscilloscope, DMM, 3D Printer, Electronic Components, Computer Hardware, FPGAs, Embedded Firmware, PCB Design

Work Experience

NYC Department for the Aging

Special Initiatives Intern

New York, NY

July 2025 - August 2025

- Assisted in packing 70,000 coupons with a net worth of \$1.75M under NYC's Farmer's Market Nutrition Program for older adults
- Mastered various types of Microsoft 365 services (Word, Excel, PowerPoint, Sharepoint, Outlook, etc.)
- Managed Excel spreadsheets of coupon orders across over 400 older adult centers participating in the program

Stony Brook College of Engineering & Applied Sciences

Undergraduate Teaching Assistant

Stony Brook, NY

January 2025 - May 2025

- Served as a teaching assistant under Prof. Kenneth Short for the design course Digital Systems Design Using VHDL & PLDs
- Completed laboratory assignments in a digital design laboratory to verify VHDL simulations on a variety of target CPLDs and FPGAs
- Operated various CPLDs and FPGAs during lab sessions and gave students credit when they successfully complete tasks
- Held office hours for an hour every week to give students additional assistance in coursework

One World Observatory

Observatory Ambassador

New York, NY

July 2024 - August 2024

- Directed guests to appropriate areas throughout the main lobby, elevators, theater, and observation deck
 - Handled the controls of the high-speed elevators from the ground and observation deck floors
 - Managed the video presentation at the See Forever Theater on the 102nd floor
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Projects

Customized RISC-V Processor on an FPGA

Senior Design Project

September 2025 - May 2026

- Customized a 5-stage RISC-V processor with a branch prediction algorithm intended to reduce occurrences of pipeline stalls
- Designed each module of the RISC-V processor (Register Files, ALU, Hazard Detection Unit, etc.) using the SystemVerilog HDL
- Forwarded RV32I integer operations and RV32F floating point operations into the RISC-V pipeline to verify arithmetic computations
- Applied Universal Verification Methodology to ensure all modules of the RISC-V processor are functioning as intended
- Implemented the design on an AMD / Xilinx Zynq SoC with programmability that is similar to an ARM-based processor and an FPGA

Pipelined SIMD Multimedia Unit

Computer Architecture

March 2025 - May 2025

- Built a 4-stage pipelined multimedia unit with similar architectures to the Intel SSE and the design of the Sony Cell SPU processor
- Wrote VHDL design descriptions for modules in each stage (ie. instruction buffer, register file, data forwarding unit, ALU)
- Verified the functionality of the ALU at the "execute" stage following the operation of the data forwarding unit
- Used the C language to convert a text file of Assembly Language instructions to binary format for the instruction buffer
- Linked all modules together using a structural implementation that includes pipeline registers sensitive to a system clock